

Query Optimization: Estimating Statistics

1 Statistical Information for Cost Estimation

The cost of a query depends on the size of the intermediate results. The database catalog stores metadata to facilitate these estimates.

Core Statistics

- n_r : Number of tuples in relation r .
- b_r : Number of blocks containing tuples of r .
- l_r : Size of a single tuple of r (in bytes).
- f_r : Blocking factor (number of tuples per block).
- $V(A, r)$: Number of distinct values for attribute A in r ; equivalent to $|\Pi_A(r)|$.

Physical Assumption: $b_r = \lceil n_r / f_r \rceil$.

1.1 Histograms

Histograms provide a refined view of data distribution.

- **Equi-width Histograms:** Divide attribute values into equal-sized ranges.
- **Equi-depth Histograms:** Ranges are adjusted so each bucket contains roughly the same number of tuples (e.g., boundaries like 4, 8, 14, 19).
- **Frequent Values:** Many systems store the n most frequent values and their counts separately to improve accuracy.

Maintenance

Statistics are often computed via random sampling. Since updates are periodic (manual ANALYZE or automatic after certain thresholds), statistics may be slightly out of date.

2 Selection Size Estimation

2.1 Equality Predicates ($\sigma_{A=v}(r)$)

- **General Case:** Estimated size $c = n_r / V(A, r)$.
- **Key Attribute:** If A is a key, the size estimate is 1.

2.2 Comparison Predicates ($\sigma_{A \leq v}(r)$)

Let c be the estimated number of tuples:

- If $v < \min(A, r)$, then $c = 0$.
- If $v \geq \max(A, r)$, then $c = n_r$.
- Otherwise, $c = n_r \cdot \frac{v - \min(A, r)}{\max(A, r) - \min(A, r)}$.

Fallback: In the absence of min/max info, c is typically assumed to be $n_r/2$.

2.3 Complex Selections (Selectivity)

Selectivity of condition θ_i is s_i/n_r , where s_i is the number of satisfying tuples.

- **Conjunction** ($\theta_1 \wedge \dots \wedge \theta_n$): $n_r \cdot \prod (s_i/n_r)$ (assuming independence).
- **Disjunction** ($\theta_1 \vee \dots \vee \theta_n$): $n_r \cdot [1 - \prod (1 - s_i/n_r)]$.
- **Negation** ($\neg\theta$): $n_r - \text{size}(\sigma_\theta(r))$.

3 Join Size Estimation

Running Example: $\text{student} \bowtie \text{takes}$

Parameters:

- $n_{\text{student}} = 5,000$, $n_{\text{takes}} = 10,000$.
- $f_{\text{student}} = 50 \implies b_{\text{student}} = 100$.
- $f_{\text{takes}} = 25 \implies b_{\text{takes}} = 400$.
- $V(ID, \text{takes}) = 2,500$ (Average of 4 courses per active student).
- $V(ID, \text{student}) = 5,000$ (Primary Key). ID in takes is a foreign key.

3.1 Join Size Formulas

- **Cartesian Product** ($r \times s$): $n_r \cdot n_s$ tuples.
- **Disjoint Schemas** ($R \cap S = \emptyset$): Same as Cartesian.
- **Key/Foreign Key Relationship:**
 - If $R \cap S$ is a key for R , then $\text{size} \leq n_s$.
 - If it's a Foreign Key in S referencing R , then $\text{size} = n_s$. (Example: $\text{student} \bowtie \text{takes} = 10,000$).
- **General Case** ($R \cap S = \{A\}$): Pick the lower of:

$$\frac{n_r \cdot n_s}{V(A, s)} \quad \text{or} \quad \frac{n_r \cdot n_s}{V(A, r)}$$

4 Estimating Other Operations

- **Projection** ($\Pi_A(r)$): Estimated size is $V(A, r)$.
- **Aggregation** ($\gamma_A(r)$): Size is $V(G, r)$.
- **Set Operations:**
 - **Same relation:** Rewrite as disjunction ($\cup$) or conjunction ($\cap$).
 - **Different relations:** Union ($n_r + n_s$), Intersection ($\min(n_r, n_s)$), Difference (n_r).
- **Outer Joins:**
 - Left (r LEFT OUTER JOIN s): $\text{size}(r \bowtie s) + n_r$.
 - Full (r FULL OUTER JOIN s): $\text{size}(r \bowtie s) + n_r + n_s$.

5 Number of Distinct Values

5.1 Selections ($\sigma_\theta(r)$)

- $A = v \implies V = 1$.
- $A \in \{v_1, \dots, v_n\} \implies V = |\{v_1, \dots, v_n\}|$.
- $A \text{ op } v \implies V = V(A, r) \cdot \text{selectivity}$.
- General Case: $\min(V(A, r), n_{\sigma_\theta(r)})$.

5.2 Joins ($r \bowtie s$)

Let A be attributes from r (A_1) and s (A_2).

- If $A \subseteq R$: $\min(V(A, r), n_{\bowtie})$.
- Mixed Attributes: $\min(V(A_1, r) \cdot V(A_2 - A_1, s), V(A_1 - A_2, r) \cdot V(A_2, s), n_{\bowtie})$.

Aggregates

For $\min(A)$ and $\max(A)$, $V \approx \min(V(A, r), V(G, r))$. For other aggregates, we often assume all values are distinct ($V = V(G, r)$).